

Lec: Root Cause Analysis - I (RCA)

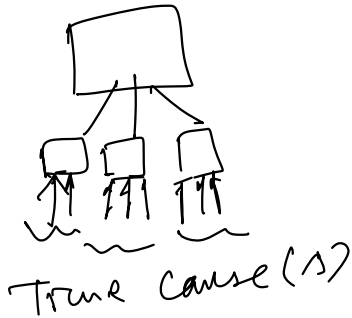
Agenda:

- ✓ ① Myntra case study
- ✓ ② Root cause Analysis (RCA)
- ✓ ③ Life cycle of RCA
- ✓ ④ Assumptions
- ⇒ ⑤ User Journey funnel
- ✓ ⑥ CRIED
- ⑦ Payment related issues
- ⑧ conclusion.

#RCA: Need.

↳ Root Cause ?

RCA: structure, methodical process
to identify the fundamental
reason behind a problem;
not just the surface level
symptoms.



3 core goals of RCA

- ✓ ① Identify :
 - ✓ ② Understand :
 - ✓ ③ Prevent :
-

RCA in action:

Q1. What is the problem?

Q2. Why did it happen?

↳ 5 whys?

↳ why did TE crash?

↳ Ran out of memory.

↳ why out of memory?

↳ Too many users
logged at once

↳ why so many users?

↳ Launch premiere increased the traffic.

↑
↳ why no capacity prep?

↳ Nissan India
↳ B-MPV (5+2)
(under 102)

- ↳ server capacity increased,
 - ↳ Why no prep?
 - ↳ The team didn't plan for capacity.
-

② Reconstruct Timeline :

slow delivery

Day 1: Order placed

Day 2: Warehouse delay

Day 3: Delivery truck broke down

Day 4: Package delivered late

Q3. What needs to be done so that it never occurs again?

↳ Preventive measure

How would a business choose what metric it should focus on?

① Align with the big goals:

<u>Goal</u>	<u>Metric</u>
① Reduce churn	churn rate
② Increase engagement	Engagement rate

② Context A communication :

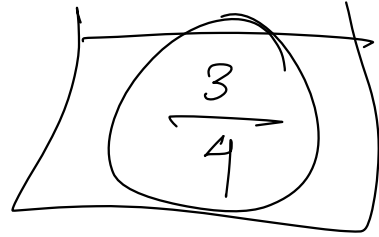
<u>pos metric</u>	<u>neg negative</u>
<u>conversion rate ↑</u>	<u>error rate ↑</u>

③ Actionability :

(cart abandonment rate)
➔



cart addition count



④

Track both metrics

(metric) vs (Antonym metric)

churn rate
profit

v1
v4

Retention Rate
loss

⑤ Stakeholder preferences:

Lifecycle of ^{RC} Analysis:

① Clarify

② Rule out

③ Internal factors (data)

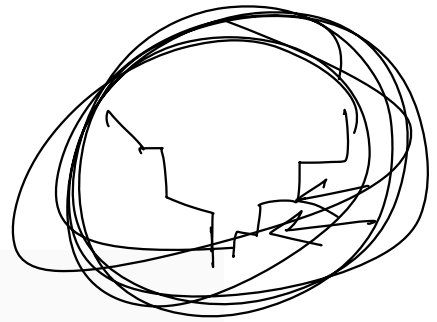
④ External factors (data)

CRIED
↓

TROPICS

Myntra Case Study !

$$\frac{5-3}{5} = \frac{2}{5} \times 100 = 40\%$$



Problem Statement:

Myntra has observed a decline from 5% to 3% (i.e., 2%) in the number of confirmed orders over the last 3 days.

You have to perform a Root Cause Analysis and help us diagnose the issue.

Assumptions :

- ① Definition of cart addition
- ② Definition of confirmed orders
- ③ Define a session.

④ orders per session $\hat{=}$ # of orders placed
in a day

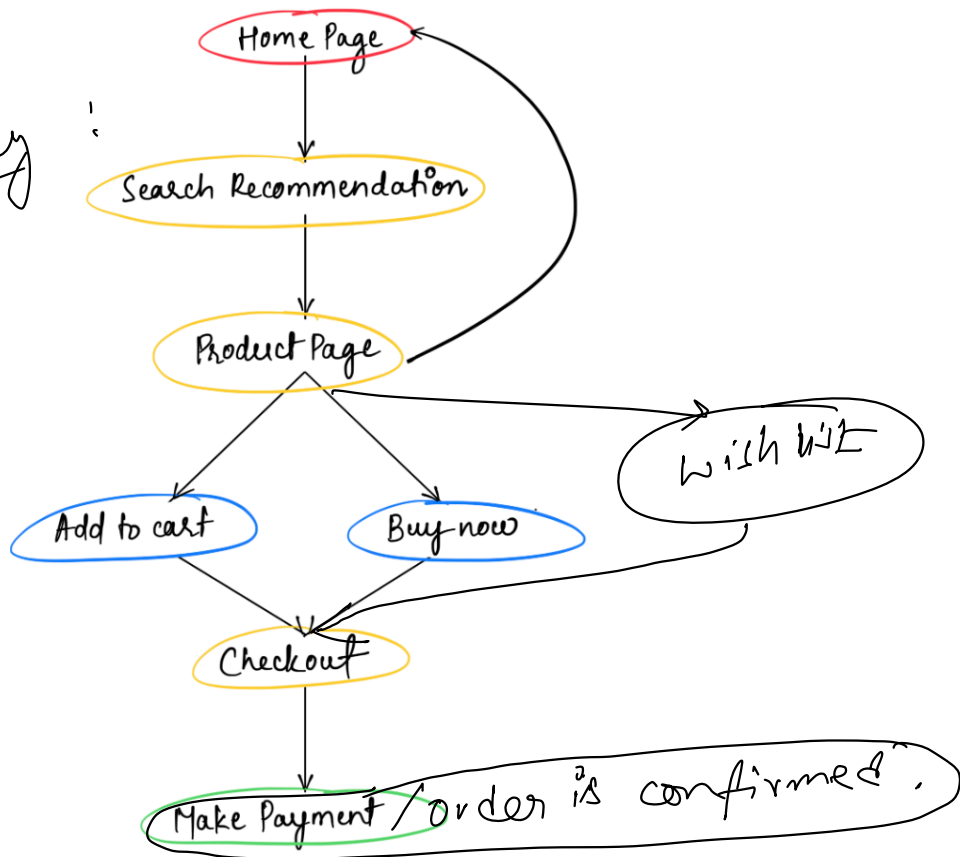


Total # of sessions
accessed that day,

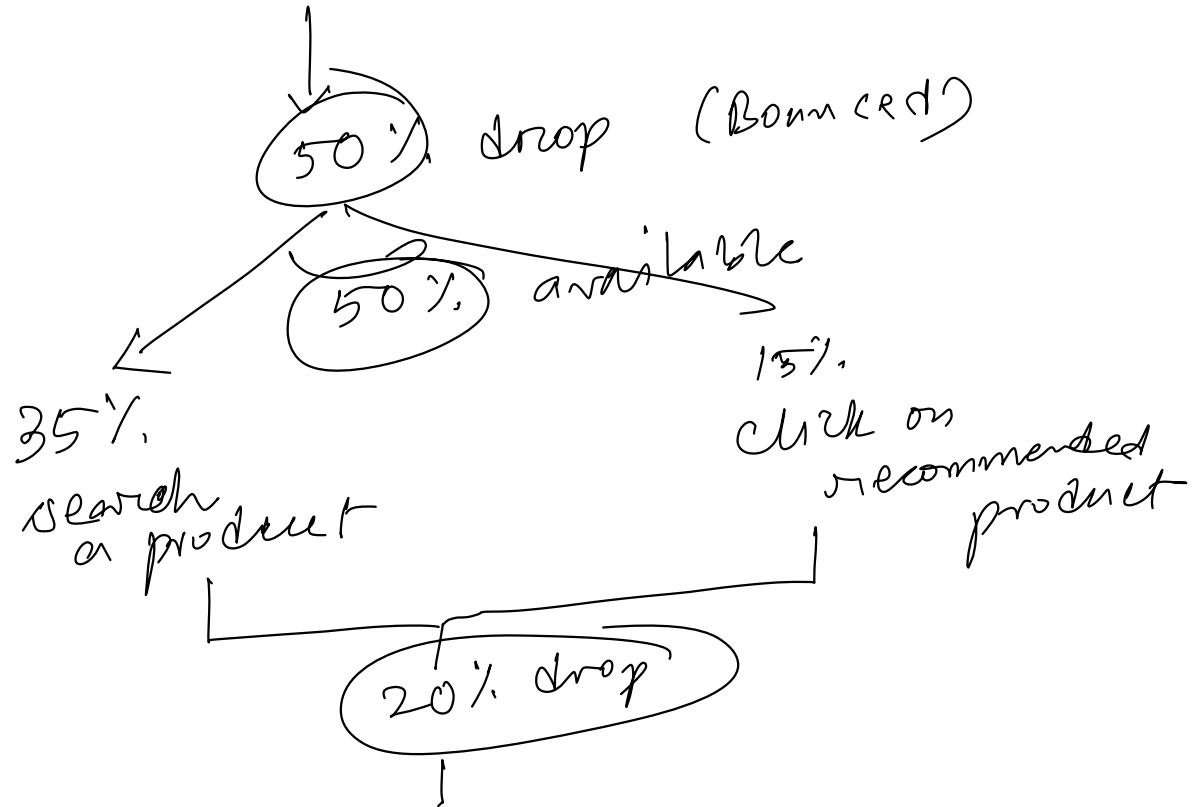
⑤ Bounce session

Bounce rate $\hat{=}$ $\frac{\text{\# of bounce session in a day}}{\text{Total \# of sessions accessed that day,}}$

User Funnel Journey

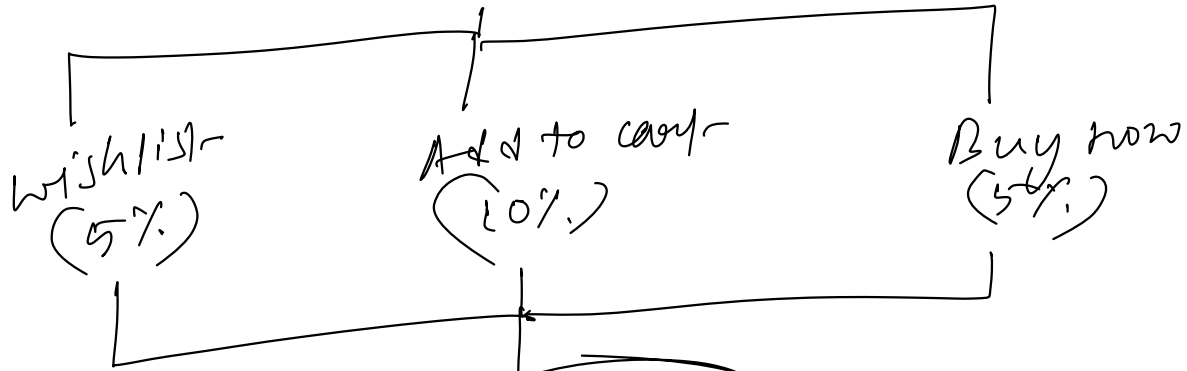


Homepage (100 unique session ids)



product page (30%) available

10% drop (20% remains)



10% drop

checkout (10% available)

↓
5% drop

↓
order confirmation (5% ~~over~~ variable)
(paid on COD) ↓ 2%
3%

External factors

↳

~~_____~~

Diagnose the Root Cause;

Q.

Q.

Q.

Q.

Payment related Issue

